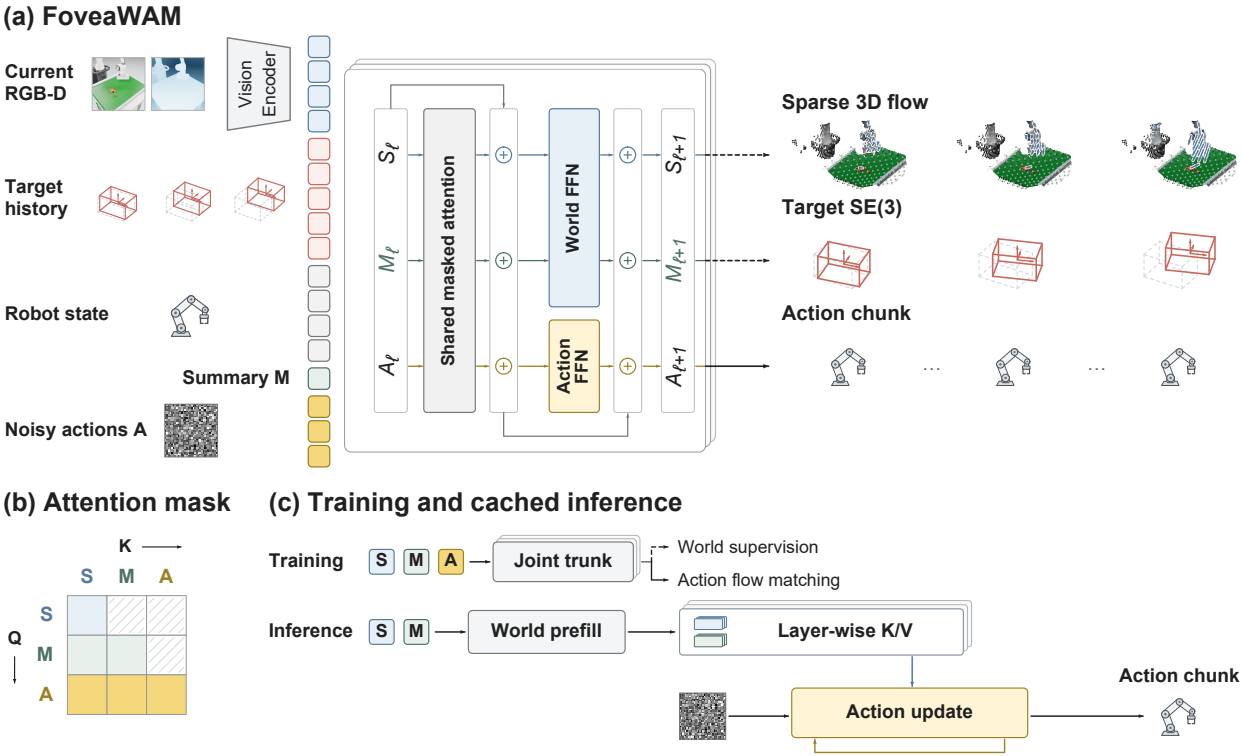
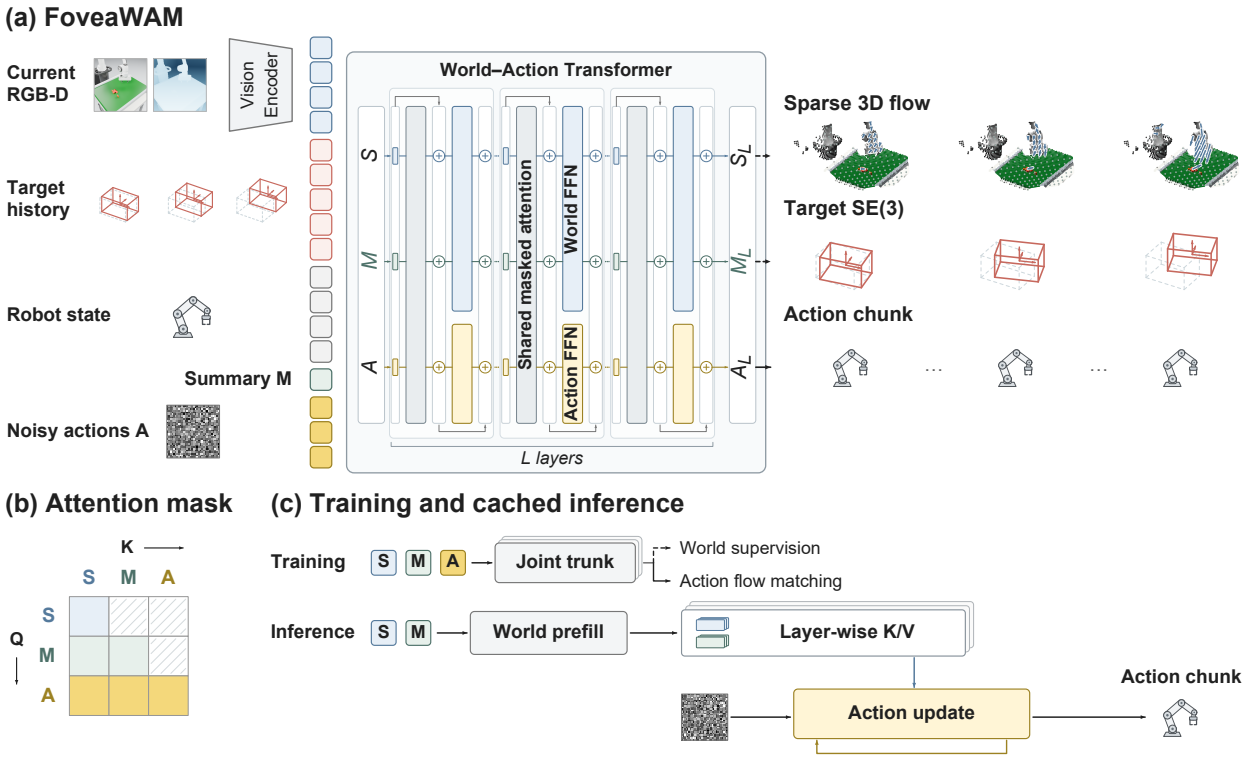
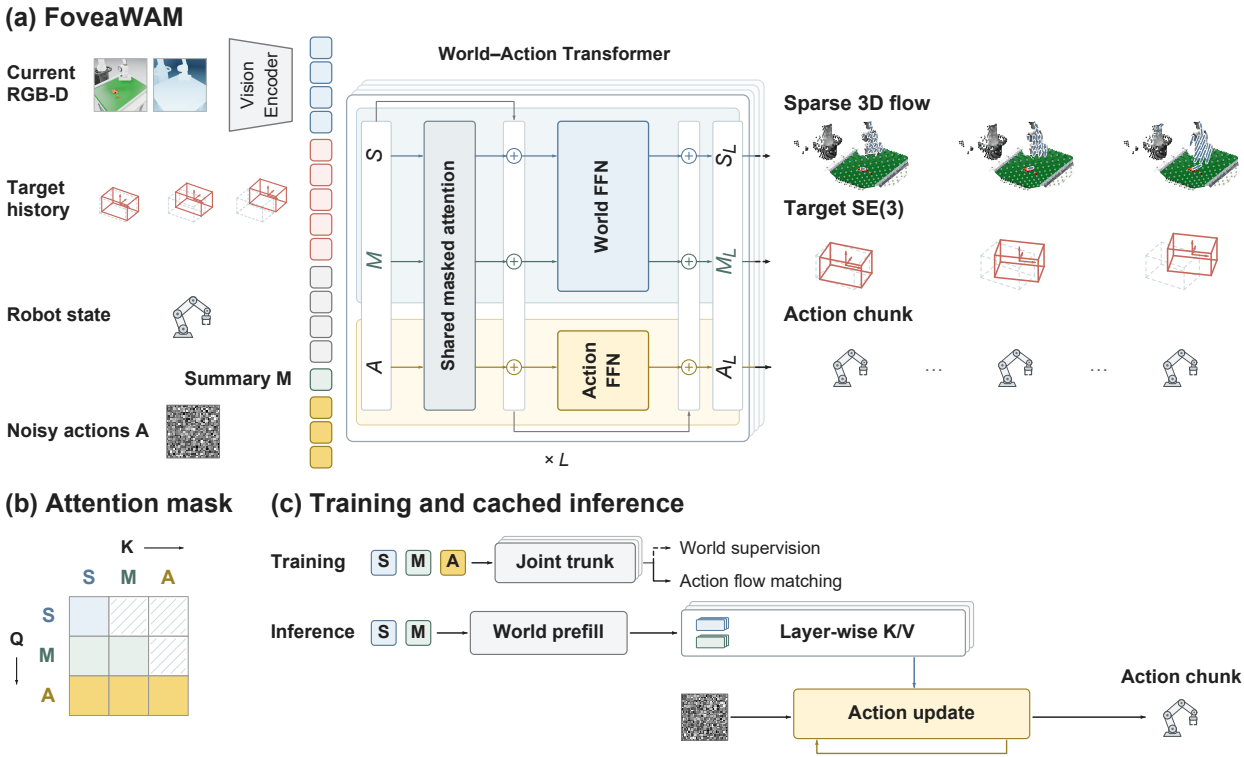

(a) FoveaWAM
Current RGB-D
Vision Encoder
Target history
Robot state
Summary M
Noisy actions A
S_ℓ
M_ℓ
A_ℓ
Shared masked attention
World FFN
Action FFN
$S_{\ell+1}$
$M_{\ell+1}$
$A_{\ell+1}$
Sparse 3D flow
Target SE(3)
Action chunk
(b) Attention mask
K
S
M
A
Q
S
M
A
(c) Training and cached inference
Training
S
M
A
Joint trunk
World supervision
Action flow matching
Inference
S
M
World prefill
Layer-wise K/V
Action update
Action chunk


(a) FoveaWAM
Current RGB-D
Vision Encoder
Target history
Robot state
Summary M
Noisy actions A
World–Action Transformer
S
M
A
Shared masked attention
World FFN
Action FFN
S_L
M_L
A_L
L layers
Sparse 3D flow
Target SE(3)
Action chunk
(b) Attention mask
K
S M A
Q
S
M
A
(c) Training and cached inference
Training
S M A
Joint trunk
World supervision
Action flow matching
Inference
S M
World prefill
Layer-wise K/V
Action update
Action chunk


(a) FoveaWAM
Current RGB-D
Vision Encoder
Target history
Robot state
Summary M
Noisy actions A
World–Action Transformer
S
M
A
Shared masked attention
World FFN
Action FFN
S_L
M_L
A_L
× L
Sparse 3D flow
Target SE(3)
Action chunk
(b) Attention mask
K
S M A
Q
S
M
A
(c) Training and cached inference
Training
S M A
Joint trunk
World supervision
Action flow matching
Inference
S M
World prefill
Layer-wise K/V
Action update
Action chunk